

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

GENUS PLC,
Petitioner,

v.

INGURAN, LLC,
Patent Owner.

IPR2021-01027
Patent 10,542,734 B2

Before ULRIKE W. JENKS, SUSAN L. C. MITCHELL, and
ROBERT A. POLLOCK, Administrative Patent Judges.

JENKS, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision
Determining All Claims Unpatentable
35 U.S.C. § 318

I. INTRODUCTION

This is a Final Written Decision in an *inter partes* review of claims 1–5 (“the challenged claims”) of U.S. Patent No. 10,542,734 B2 (Ex. 1001, “the ’734 patent”). We have jurisdiction under 35 U.S.C. § 6, and enter this Decision pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73. For the reasons set forth below, we determine that Genus plc (“Petitioner”) has shown, by a preponderance of the evidence, that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e).

A. *Procedural History*

Genus plc (“Petitioner”) filed a Petition for an *inter partes* review of the challenged claims under 35 U.S.C. § 311. Paper 3 (“Pet.”). Petitioner supported the Petition with the Declaration of Dr. Christine Baes. Ex. 1002. Inguran, LLC (“Patent Owner”) filed a Preliminary Response to the Petition. Paper 7 (“Prelim. Resp.”).

On December 14, 2021, pursuant to 35 U.S.C. § 314(a), we instituted trial (“Decision” or “Dec.”) to determine whether any challenged claim of the ’734 patent is unpatentable based on the grounds raised in the Petition:

Claim(s) Challenged	Basis¹	References
1–5	§ 102(b) ²	Dekkers

¹ The relevant sections of the Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112–29, 125 Stat. 284 (Sept. 16, 2011), took effect on March 16, 2013. Because the priority application from which the ’734 patent issued was filed before that date, our citations to Title 35 are to its pre-AIA version.

² Petitioner contends that claims 1–5 are anticipated by Dekkers, as evidenced by either Maxwell or Garner (claims 1–4), or Kim (claim 5). Pet. 26–41; Pet. Reply 8–17.

Claim(s) Challenged	Basis ¹	References
1–5	§ 103(a)	Dekkers, Maxwell, Garner, Kim
1–5	§ 103(a)	Dekkers, Okere
1–5	§ 103(a)	Buttram, Maxwell, Kim ³

Patent Owner filed a Patent Owner Response to the Petition. Paper 12 (“PO Resp.”). Patent Owner supported the Response with a Supplemental Declaration of Dr. Hein van der Steen. Ex. 2005. *See* PO Resp., iv (Exhibit List). Petitioner filed a Reply to the Patent Owner Response. Paper 14 (“Pet. Reply”). Petitioner supported the Reply with a Supplemental Declaration from Dr. Christine Baes. Ex. 1097. Patent Owner filed a Sur-Reply to Petitioner’s Reply. Paper 18 (“PO Sur-Reply”).

On September 13, 2022, the parties presented arguments at an oral hearing. Paper 21. The hearing transcript has been entered in the record. Paper 24 (“Tr.”).

For the reasons set forth below, we determine that Petitioner has shown by a preponderance of the evidence that claims 1–5 of the ’734 patent are unpatentable.

B. Real Parties-in-Interest

Petitioner identifies itself and ABS Global, Inc., as real parties-in-interest to this proceeding. Pet. 3. Patent Owner identifies itself as real party-in-interest. Paper 5, 1.

³ In the table identifying ground III, Petitioner does not list Kim. Pet. 4. However, in the arguments provided for this ground in regard to claim 5, Petitioner includes Kim. Pet. 48.

C. Related Proceedings

The parties state that they are not aware of any other judicial or administrative matter that would affect, or be affected by, a decision in this proceeding. Pet. 3; Paper 5, 1. The parties indicate that pending U.S. Application No. 16/708,007 claims priority to the '734 patent. *Id.*

D. The '734 Patent (Ex. 1001)

The '734 patent issued on Jan. 28, 2020 from Application No. 16/248,569 (“the '569 application”), filed Jan. 15, 2019. The '734 patent is titled “Methods for Increasing Genetic Progress in a Line or Breed of Swine Using Sex-selected Sperm Cells.” Ex. 1001, code (54). The '734 patent discloses “methods of increasing the genetic progress of a line, breed or herd of swine through the use of sex-selected sperm cells in artificial insemination techniques.” *Id.* at 1:17–20. As background, the '734 patent indicates that there is “a need in the swine industry to increase the rate of desirable genetic change in lines and breeds as well as to lower operational costs on breeding and commercial swine farms.” *Id.* at 1:27–30. The '734 patent purports to achieve these goals by allowing an operator of a breeding or commercial farm to “select the sex of offspring swine using sex-selected sperm cell samples and/or by allowing the use of far fewer genetically superior boars through the use of reduced sperm cell doses for artificial insemination procedures.” *Id.* at 1:30–35.

A “line” or “breed” is “a group of animals having a common origin and similar identifying characteristics.” *Id.* at 3:3–5. The '734 patent describes that “any desirable genotypic or phenotypic trait can be used to construct a selection index for a line or breed of swine” and that such

phenotypic traits include “feed efficiency,^[4] average daily gain, carcass lean, carcass quality, fertility, litter size and milk production.” *Id.* at 3:18–23. The rate of desirable genetic change of a herd or line can be increased by selecting the sex of the offspring in a line. *Id.* at 7:28–35. A genetic change, or a “genetic progress” means that for a given desired trait or a given undesirable trait, “the individuals in the successive generation will express the desired trait more strongly than the previous generation” or “the individuals in the successive generation will express the [undesirable] trait less strongly than the previous generation.” *Id.* at 7:30–36.

The ’734 patent discloses that swine production “can be represented by a multilevel pyramid, with certain offspring at each level used in the next lower level for breeding.” *Id.* at 9:41–43. The top level of the pyramid is the “nucleus herd” and the next levels from top to bottom are the “daughter nucleus herd,” the “multiplier herd” and finally the “commercial farm,” respectively. *Id.* at 9:43–46. In such a pyramid, the nucleus herd “is typically comprised of 12 to 15 lines, although any number of lines may be represented, with each line comprising a number of desirable traits.” *Id.* at 9:47–49. The daughter nucleus level is a “line multiplication” level in which “sows from a line are inseminated with boar semen derived from the nucleus herd.” *Id.* at 10:17–19. In the next lower level of the multiplier herd level, “the resulting sows are typically crossed with boars of a different line.” *Id.* at 10:19–21. Here, a multiplier herd “is typically either a gilt multiplier or boar

⁴ Feed efficiency is “a measure of an animal’s efficiency in converting feed mass into increased body mass (also known as feed conversion or feed to gain ratio)” and average daily gain is “the average daily weight gain for an animal.” *Id.* at 9:5–9.

multiplier.” *Id.* at 10:21–22. A gilt is a young female pig that has not yet fallowed. *Id.* at 6:20–21. Gilt multipliers are then used to “produce parent gilts” and boar multipliers are then used to “produce parent boars.” *Id.* at 10:22–23. “Parent gilts are sent to commercial farms to replace old or dead gilts and sows, and parent boars are sent to boar studs to produce the sperm cells for use in artificial insemination.” *Id.* at 10:23–26.

According to the ’734 patent, its method of increasing the profitability of a swine herd or farm includes “collecting a semen sample from a boar” and “sorting said semen sample into at least two subpopulations of sperm cells, wherein at least 80% of a first subpopulation bears (i) X-chromosomes if the female pig results in a higher net income per pig, or (ii) Y-chromosomes if the male pig results in a higher net income per pig.” *Id.* at 6:6–17. A sow is then inseminated with sperm cells from the first subpopulation to produce offspring. *Id.* at 6:17–19. Further, an embryo or a zygote is selected “based on the presence or absence of a genetic marker” and the “genetic marker can be a marker for, or associated with, for example, any of the swine traits disclosed herein.” *Id.* at 2:30–38.

E. Illustrative Claim

Claim 1 of the ’734 patent is illustrative and reproduced below:

1. [preamble] A method of increasing the genetic progress of a line or breed of swine within a genetic nucleus comprising the steps of:

[a] providing a boar from a genetic nucleus and a sow from a genetic nucleus, wherein said boar and said sow are from a line or breed within said genetic nucleus and wherein said line or breed of said boar and said sow are the same line or breed;

[b] collecting a semen sample from said boar;

- [c] sorting said semen sample into at least two subpopulations of sperm cells, wherein at least 80% of a first subpopulation bears X-chromosomes or Y-chromosomes;
- [d] inseminating said sow with sperm cells from said first subpopulation to produce offspring from said sow;
- [e] genotyping said offspring to identify one or more genetic markers;
- [f] selecting one or more of said offspring as a parent in the line or breed in the genetic nucleus based on the one or more identified genetic markers.

Ex. 1001, 25:8–26:8 (lettering and bracketing added for reference convenience).

II. ANALYSIS

A. Person of Ordinary Skill in the Art

Petitioner asserts that a person of ordinary skill in the art would have had at least “an undergraduate or graduate degree in genetics, genomics, and/or animal breeding, along with several years of relevant applied research and/or industry work experience in the same fields, including in swine” and “would have been well-acquainted with methods of commercial breeding of lines and breeds within a genetic nucleus for the purpose of promoting genetic progress, the use of sex-selected semen in breeding schemes, including those within the genetic nucleus, genetic evaluation of livestock for desirable and undesirable traits, and the use of genetic markers, *e.g.*, SNPs, and various other genotype data, to guide selection strategies to improve desirable traits and/or minimize undesirable traits.” Pet. 13 (citing Ex. 1002 ¶ 22).

Patent Owner agrees with Petitioner’s first portion of the definition, specifically, that a person of ordinary skill in the art would have had at least

“an undergraduate or graduate degree in genetics, genomics, and/or animal breeding, along with several years of relevant applied research and/or industry work experience in the same fields, including in swine.” PO Resp. 16. Patent Owner, however, objects to the inclusion of the second portion of Petitioner’s definition because it is “improperly injecting a purported knowledge of the claimed invention into the definition of a” person of ordinary skill in the art. *Id.* at 17 (citing *Arendi S.A.R.L. v. Apple Inc.*, 832 F.3d 1355, 1362 (Fed. Cir. 2016); *In re Sang Su Lee*, 277 F.3d 1338, 1345 (Fed. Cir. 2002)); Sur-reply 2–4. Petitioner responds, that its proposed definition reflects the level of skill in the art as supported by the various cited prior art. Pet. Reply. 5.

Having considered the arguments and evidence, we agree with Patent Owner that Petitioner’s definition of one of ordinary skill in the art unnecessarily recites specific knowledge, whereas the art itself demonstrates the level of skill in the art at the time of the invention. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (explaining that specific findings regarding ordinary skill level are not required “where the prior art itself reflects an appropriate level and a need for testimony is not shown”) (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985)). Accordingly, we agree with Patent Owner and determine that one of ordinary skill in the art is a person having at least “an undergraduate or graduate degree in genetics, genomics, and/or animal breeding, along with several years of relevant applied research and/or industry work experience in the same fields, including in swine” and this definition is reasonably consistent with the ’734 patent and the prior art of record.

B. Principles of Law

To establish anticipation under 35 U.S.C. § 102, each and every element in a claim, arranged as recited in the claim, must be found in a single prior art reference. *See Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008); *Karsten Mfg. Corp. v. Cleveland Golf Co.*, 242 F.3d 1376, 1383 (Fed. Cir. 2001). Although the elements must be arranged or combined in the same way as in the claim, “the reference need not satisfy an *ipsissimis verbis* test,” i.e., identity of terminology is not required. *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009) (accord *In re Bond*, 910 F.2d 831, 832–33 (Fed. Cir. 1990)).

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time of the invention to a person having ordinary skill in the art. *KSR Int’l Co. v. Teleflex, Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including “the scope and content of the prior art”; “differences between the prior art and the claims at issue”; and “the level of ordinary skill in the pertinent art.” *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). *Graham* also identifies secondary considerations of nonobviousness as part of the inquiry but no such evidence has been presented during this proceeding.

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled

artisan would have had a reasonable expectation of success in doing so.”
Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd., 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation omitted); *see KSR*, 550 U.S. at 418 (for an obviousness analysis, “it can be important to identify a reason that would have prompted a person of ordinary skill in the relevant field to combine the elements in the way the claimed new invention does”). Further, an assertion of obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418 (quoting *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)); *In re NuVasive, Inc.*, 842 F.3d 1376, 1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’” (citation omitted)).

C. Claim Construction

In an *inter partes* review for a petition filed on or after November 13, 2018, “claim[s] of a patent . . . shall be construed using the same claim construction standard that would be used to construe the claim[s] in a civil action under 35 U.S.C. [§] 282(b), including construing the claim[s] in accordance with the ordinary and customary meaning of such claim[s] as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” 37 C.F.R. § 42.100(b); *see* Changes to the Claim Construction Standard for Interpreting Claims in Trial Proceedings Before the Patent Trial and Appeal Board, 83 Fed. Reg. 51,340 (Oct. 11, 2018) (codified at 37 C.F.R. § 42.100(b) (2019)) (amending 37 C.F.R. § 42.100(b) effective November 13, 2018). A term’s ordinary and customary meaning “is its meaning to the ordinary artisan after reading the entire patent.”

Phillips v. AWH Corp., 415 F.3d 1303, 1321 (Fed. Cir. 2005) (en banc). Nevertheless, “it is always necessary to review the specification to determine whether the inventor has used any terms in a manner inconsistent with their ordinary meaning,” because “[t]he specification acts as a dictionary when it expressly defines terms used in the claims or when it defines terms by implication.” *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996).

Both Petitioner and Patent Owner agree that the preamble is limiting. *See* Pet 27 (“The preamble to claim 1 of the ’734 patent is limiting in its recitation “increasing the genetic progress of a line or breed of swine within a genetic nucleus.”); PO Resp. 20.

Petitioner requests construction of the terms “genetic progress,” “line or breed,” “genetic nucleus,” “feed efficiency,” “average daily gain,” and “carcass lean.” Pet. 9–12. The terms “genetic progress,” “line or breed,” and “genetic nucleus” are found in the independent claim 1. The term “line or breed” is also found in some of the dependent claims. While the terms “feed efficiency,” “average daily gain,” and “carcass lean” are terms found in dependent claim 5.

Patent Owner only provides a separate construction of “genetic nucleus” PO Resp. 20–24.

1. genetic nucleus

According to Petitioner, “[t]he ’734 patent discloses that ‘[s]wine production can be represented by a multilevel pyramid, with certain offspring at each level used in the next lower level for breeding,’ that the ‘top level of the pyramid is the nucleus herd,’ and that ‘any number of lines may be represented’ in a nucleus herd.” Pet. 10 (citing Ex. 1001, 9:41–49)

(second alteration in original). Petitioner contends that this provides an incomplete description of genetic nucleus because “commercial breeders must also use offspring of the genetic nucleus for breeding within the genetic nucleus itself.” *Id.* (citing Ex. 1004, 410 (“Most genetic improvement in pigs is based on selection within pure breeds or lines.”), 417 (Fig. 16.4); Ex. 1002 ¶ 32). Petitioner proposes that “genetic nucleus” means “an animal herd comprising one or more lines and positioned at the top of a multi-level pyramidal hierarchy of herds in a commercial herd breeding structure, wherein certain offspring of the herd are used for breeding within the herd and certain offspring are used for breeding in the next lower level of the pyramid.” *Id.* at 10 (citing Ex. 1002 ¶ 32).

Patent Owner agrees with the first portion of Petitioner’s interpretation, but disagrees on the remainder. Patent Owner argues that one of ordinary skill in the art would not find the additional language “wherein certain offspring of the herd are used for breeding within the herd” in Petitioner’s interpretation accurate in describing a genetic nucleus. PO Resp. 21 (citing Ex. 2001 ¶ 21). Patent Owner contends “that the plain and ordinary meaning of ‘genetic nucleus’ is ‘an animal herd comprising one or more lines and positioned at the top of a multi-level pyramidal hierarchy of herds in a commercial herd breeding structure.’” *Id.* (citing Ex. 2005 ¶¶ 31–33 (“The ’734 Patent does not specify this additional requirement as part of the meaning of ‘genetic nucleus,’ nor would the additional language be accurate in describing every genetic nucleus.”))).

Nucleus herd and genetic nucleus are understood to be synonymous terms. *See* Tr. 8:22–9:10; Ex. 1002 ¶ 40 (“At the peak of the pyramid is the *nucleus herd*, also known as the *genetic nucleus*. This is the herd in which

genetic improvements are implemented, such as by the isolation or introduction of desirable traits.”). The ’734 specification describes pig breeding as a pyramid with the “nucleus herd” at the top of the pyramid and commercial farms at the bottom. Ex. 1001, 9:43–44. According to the ’734 specification, “[a] nucleus herd is typically comprised of 12 to 15 lines, although any number of lines may be represented, with each line comprising a number of desirable traits. . . . *Only the offspring with the highest measured selection index are retained in the nucleus herd.*” *Id.* at 9:47–49 (emphasis added). Here, the retention of offspring with the most desirable features in the nucleus herd reasonably implies that these individuals are selected to also propagate the desirable trait within the nucleus herd.

Having considered the record as a whole, we determine that Petitioner’s position with respect to the interpretation of “genetic nucleus” is reasonably supported within the record. At oral argument, however, Petitioner acknowledged that omitting the language “wherein certain offspring of the herd are used for breeding within the herd” from the claim construction would not change whether the challenged claims read on the applied references. *See* Tr. 14:6–16:28. Because the additional language proposed by Petitioner is unnecessary in order to resolve the disputed issues in this case, we do not adopt it here. We, therefore, construe “genetic nucleus” as an animal herd comprising one or more lines and positioned at the top of a multi-level pyramidal hierarchy of herds in a commercial herd breeding structure.

2. Remaining Claim Terms

We find that it is unnecessary to expressly construe any remaining claim terms for purposes of this Decision. *See Wellman, Inc. v. Eastman*

Chem. Co., 642 F.3d 1355, 1361 (Fed. Cir. 2011) (“[C]laim terms need only be construed ‘to the extent necessary to resolve the controversy.’”) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

D. Ground 1A: Anticipation of Claims 1–5 by Dekkers, as evidenced by either Maxwell or Garner (claims 1–4), or Kim (claim 5)

Petitioner contends that claims 1–5 are unpatentable as anticipated by Dekkers, as evidenced by either Maxwell or Garner (claims 1–4), or Kim (claim 5). Pet. 26–41; Pet. Reply 8–16. Patent Owner opposes. PO Resp. 24–39; Sur-reply 10–16.

1. Overview of Dekkers (Ex. 1004)

Dekkers teaches “[g]enetic improvement in pigs.” Ex. 1004, 390. According to Dekkers, the general principles of genetic improvement includes an objective of a breeding program that produces “most improvement per unit of time.” *Id.* at 391. A “goal is then to maximize genetic change over generations or over time.” *Id.* at 392. Dekkers indicates that in most countries, pig breeding programs “operate in a three-tier pyramidal structure.” *Id.* at 393. Dekkers discloses that at the peak of the pyramid are the “nucleus breeding farms that actually generate the genetic changes.” *Id.* (emphasis omitted). Below that tier are “multiplier farms that carry out specific matings or crossbreeding for the production of large numbers of females.” *Id.* (emphasis omitted). These females are then sold, under the lower tier, to “commercial producers for piglet production and finishing to produce market pigs” for slaughterhouses. *Id.* (emphasis omitted). Variations of the tiers of the pyramid are possible. *Id.*

Dekkers further discloses that one of the most used reproduction technologies “is AI,” or artificial insemination. *Id.* at 395–96. Dekkers also discloses sperm sexing technology that allows “the production of either male or female offspring for sire or dam lines,” which can be very useful for the nucleus and multiplication tiers. *Id.* at 396. “The technique involves the separation of spermatozoa carrying either X or Y chromosomes (Maxwell *et al.*, 2004; Garner, 2006)” in which the “sorted sperm can then be used for AI or embryo transfer.” *Id.* Dekkers also discloses that molecular genetics have revealed new classes of DNA polymorphisms, such as SNPs (single nuclear polymorphisms), that allow for direct selection of “an individual’s genotype for genes or genetic markers that are associated” with a trait. *Id.* at 407. Certain types of genetic markers are applied “*for within-breed genetic improvement.*” *Id.* Dekkers discloses that genetic selection (GS) is an enhanced version of marker assisted selection (MAS) “that involves the selection of animals for breeding on the basis of their genotype for tens of thousands of ‘random’ SNPs that cover the genome.” *Id.* at 414.

Figure 16.4 of Dekkers is reproduced below.

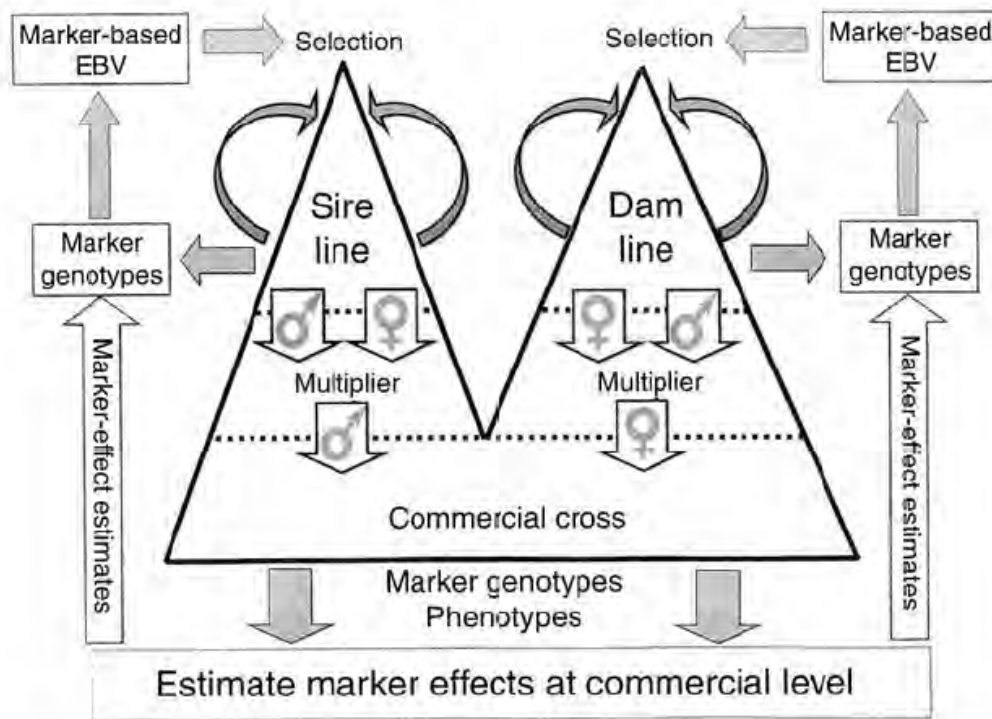


Figure 16.4 shows “[s]election of pure lines for commercial crossbred performance based on estimates of marker effects estimated on crossbreds. EBV, estimated breeding value. Based on Dekkers (2007b).” *Id.* at 417. A sire line and a dam line, each at the top of a pyramid, are used to produce a female and a male offspring for the multiplier tier, which are then used for “Commercial cross,” wherein the resulting progeny are assessed for “Marker genotypes” and “Phenotypes,” as indicated by the text below the pyramids. Estimated breeding value (EBV) is determined based on marker genotypes and an offspring is selected to be a parent as shown from the arrows surrounding the pyramids. *Id.* Dekkers states that the “nucleus breeding farms typically maintain specific breeds or breeding lines” and breeding companies “further line breeding within the crossbreds for several generations to combine and stabilize the desirable characteristics.” *Id.* at

394. As an example of genetic markers and genes used in swine, Dekkers discloses that the *MC4R* gene targets the melanocortin-4 receptor which affects the traits of lean growth, fatness, and feed intake. *Id.* at 413, Table 16.6. Thus, genetic improvement within sire or dam lines can be “targeted towards improvement in specific characteristics that are important to the producers and consumers.” *Id.* at 394.

2. Analysis

Petitioner asserts that Dekkers “discloses methods of increasing genetic progress in swine. . . . Dekkers explains that in pig breeding programs, selection response depends on, *inter alia*, selection accuracy, selection intensity or standardized selection differential, and generational interval.” Pet. 27–28 (citing Ex. 1004, 391; Ex. 1002 ¶¶ 30, 55, 56, 100–103). Petitioner further asserts that “Dekkers also teaches that the disclosed methods of genetic improvement are carried out in a line or breed at the level of the genetic nucleus (aka, nucleus breeding farms).” Pet. 29 (citing Ex. 1004, 391–392, 394). Petitioner asserts that “Dekkers expressly teaches that ‘nucleus breeding farms typically maintain specific breeds or breeding lines’ where ‘specific lines are developed to advance certain traits of interest, while keeping up the main breed characteristics.’ [] This is genetic progress.” Pet. Reply 8–9 (citing Ex. 1004, 394; Ex. 1097 ¶¶ 27–28).

Petitioner contends that Dekkers states that “sperm sexing technology allow[s] for the production of either male or female offspring for sire or dam lines, and this can be very useful for the nucleus and multiplication tiers. The technique involves the separation of spermatozoa carrying either X or Y chromosomes (Maxwell et al., 2004; Garner, 2006).” Pet. 33–34 (citing Ex. 1004, 396). Petitioner’s expert testifies that the sorting methods

referenced in Dekkers, specifically, Maxwell and Garner both evidence sorting at 85-95 % accuracy, which falls within the range cited in claim 1. *See* Ex. 1002 ¶ 113; Pet. 34.

Patent Owner opposes. *See* PO Resp. 24–39; Sur-reply 10–16. Patent Owner contends that Dekkers “cannot anticipate the challenged claims because it does not explicitly or inherently disclose sex sorting sperm cells to at least an 80% purity level.” PO Resp. 24, *see also id.* at 27 (“Dekkers does not necessarily disclose sexed semen with an 80% purity.”).

Specifically, Patent Owner contends that

Petitioner does not argue that the Dekkers reference itself expressly discloses the required 80% sex skew, and Petitioner’s expert admits the required disclosure is missing. EX 2005, ¶¶ 58–63; EX 2004, 54:17-55:10. Nor does Dekkers incorporate by reference the disclosures of Maxwell or Garner. To incorporate matter by reference, “a host document must contain language clearly identifying the subject matter which is incorporated and where it is to be found; a mere reference to another application, or patent, or publication is not an incorporation of anything therein.” *Callaway Golf Co. v. Acushnet Co.*, 576 F.3d 1331, 1346 (Fed. Cir. 2009) (internal quote and citations omitted, emphasis in original).

PO Resp. 29. Patent Owner contends that Dekkers’s paragraph referencing sex sorting technology makes only a general disclosure of sex sorting technology “and is not an express disclosure of sorting at an 80% skew rate (or any other rate).” *Id.* at 26 (citing Ex. 1004, 396; Ex. 2005 ¶¶ 58–63); Ex. 2004, 54:17–55:10 (Petitioner’s expert acknowledging that “Dekkers doesn’t write 80 percent in this paragraph.”).

In response, Petitioner contends that “Maxwell and Garner—each cited in Dekkers—evidence knowledge attributable to a [person of ordinary

skill in the art] regarding porcine sex-skewed semen.” Pet. Reply 14 (citing Ex. 1097 ¶ 45). Dr. Baes testifies that “from a commercial pig production perspective, [a person of ordinary skill in the art would understand that] using semen with less than 80% sex skew would not be efficient or economical.” Pet. Reply 15 (citing Ex. 1003 ¶ 113; Ex. 1097 ¶ 46; Ex. 1095, 149 (“Sorting conditions must be optimized for each species to achieve an acceptable (90 percent or better) sperm sexing accuracy”).

We agree with Patent Owner that in an anticipation challenge under 35 U.S.C. § 102, Petitioner cannot rely upon specific disclosures in Maxwell or Gardner based on the generic reference to these publications cited Dekkers. Merely citing Maxwell or Gardner in association with sorting sperm as discussed in Dekkers is not incorporation by reference with any level of particularity. *See* PO Resp. 29 (citing *Callaway Golf Co. v. Acushnet Co.*, 576 F.3d 1331, 1346 (Fed. Cir. 2009) (To incorporate matter by reference, “a host document must contain language clearly identifying the subject matter which is incorporated and where it is to be found; a mere reference to another application, or patent, or publication is not an incorporation of anything therein.” (internal quote and citations omitted, emphasis in original)); *see also* *Cook Biotech Inc. v. Acell, Inc.*, 460 F.3d 1365, 1376 (Fed. Cir. 2006) (“To incorporate material by reference, the host document must identify with detailed particularity what specific material it incorporates and clearly indicate where that material is found in the various documents.”).

The disclosure in Dekkers relied on by Petitioner recites: “The technique involves the separation of spermatozoa carrying either X or Y chromosomes (Maxwell et al., 2004; Garner, 2006).” Ex. 1004, 396. We

note that Maxwell and Garner are review articles discussing the state of sperm sexting technology generally. *See* Ex. 1005; Ex. 1006. We agree with Patent Owner that the generic citation to these review articles as set out in Dekkers is not an incorporation by reference of any specific detail from these articles. Unlike the disclosures at issue in *Callaway* and *Cook*, Dekkers does not use language such as “reference is made to” (*Callaway*, 576 F.3d at 1346) or “the disclosure of which is expressly incorporated herein by reference” (*Cook*, 460 F.3d at 1376) in conjunction with identifying a particular portion of the cited reference that our reviewing court has held to be sufficient to indicate to one of skill in the art that the referenced material is fully incorporated in the host document. Because Dekkers does not incorporate by reference the level of sorting accuracy as required by the claims – at least 80% – the preponderance of evidence of record does not support Petitioner’s position the Dekkers anticipates claims 1–5 of the ’734 patent.

Petitioner contends that Maxwell and Garner evidence that a person of ordinary skill in the art “would have been aware of challenges associated with sex-sorted semen and that using semen with less than 80% sex skew would not be efficient or economical.” Pet. Reply 22 (citing Ex. 1002 ¶ 113). We find that this reasoning is more applicable to an obviousness-based challenge and not the present anticipation challenge.

E. Ground 1B: Obviousness of Claims 1–5 over Dekkers and either Maxwell, Garner, or Kim

Petitioner contends that claims 1–5 are unpatentable as obvious over the combination of Dekkers with either Maxwell or Garner (claims 1–4), and Kim (claim 5). Pet. 41–45. Patent Owner opposes. PO. Resp. 35–43.

1. *Overview of Maxwell (Ex. 1005)*

Maxwell teaches sex-sorting of mammalian spermatozoa “for genetic improvement of farm animals.” Ex. 1005, 79. Maxwell discloses that sex pre-selection before insemination to produce a specific sex in livestock offspring results in “faster genetic progress” for “the production of male and female crossbred lines in the pig industry.” *Id.* at 80. In sperm sexing, the “greater the difference in DNA content between X- and Y-chromosome bearing (X and Y) spermatozoa, the more efficiently the two populations can be resolved and separated by flow cytometric sorting.” *Id.* at 80–81. According to Maxwell, generally, “X and Y purities of more than 85% can be achieved in most species, depending on the rate of sorting.” *Id.* at 81.

2 *Overview of Garner (Ex. 1006)*

Garner teaches parameters that are “needed for optimization of flow cytometric sexing mammalian sperm” and updates for “the current status of sperm sexing for various species.” Ex. 1006, 943. Garner discloses that the precise measurement of “the DNA content difference between X- and Y-chromosome bearing sperm of mammals has provided an effective means of separating viable gametes carrying either the X- or Y- chromosome with an accuracy of 85–95%.” *Id.* at 944. According to Garner, “swine was the first large animal species where the feasibility of using” the technology of predetermination of sex using flow cytometry/cell sorting, was demonstrated. *Id.* at 952. Consequently, the “adoption of sperm sexing for production of founder animals with superior genetic qualities” can be used to increase the availability of animals with superior genetic traits for swine producers. *Id.*

3. *Overview of Kim (Ex. 1007)*

Kim teaches a missense variant of the porcine melanocortin-4 receptor (*MC4R*) gene that is associated with fatness, growth, and feed intake traits. Ex. 1007, 131. Identification of this missense mutation in the pig *MC4R* gene was confirmed by sequence of the PCR product. *Id.* at 133.

[G]iven the strong relationship of *MC4R* variants to leanness, growth, and feed intake, this mutation could be used immediately for marker-assisted selection . . . to develop lines of pigs to satisfy particular customer requirements. For instance, in sow lines where appetite is normally decreased after farrowing, selection for the *MC4R* allele 2 could help to improve feed intake. Furthermore, in some lines deemed to be too fat, selection for allele 1 could be employed, and in lines that were considered to be growing too slowly, allele 2 selection could be employed. Therefore, genotyping for the *MC4R* mutation in pig breeding lines will improve the selection efficiency of feed-related production traits including growth and leanness.

Id. at 134.

4. *Analysis*

Petitioner asserts that Dekkers discloses each and every element of the claimed invention. *See* Pet. 27–38, 41–45. And that the teachings of Dekkers in combination with either Maxwell or Garner renders sorting efficiencies of greater than 80% obvious. *See id.* 41–45. Petitioner provides an element-by-element comparison of the asserted art to the challenged claims. *Id.* at 29–41. Patent Owner opposes. PO Resp. 35–43. We focus our discussion below on the elements of claim 1.

Claim 1 (preamble) “A method of increasing the genetic progress of a line or breed of swine within a genetic nucleus”

Petitioner asserts that Dekkers “discloses methods of increasing genetic progress in swine. . . . Dekkers explains that in pig breeding

programs, selection response depends on, *inter alia*, selection accuracy, selection intensity or standardized selection differential, and generational interval.” Pet. 27–28 (citing Ex. 1004, 391; Ex. 1002 ¶¶ 30, 55, 56, 100–103). Petitioner further asserts that “Dekkers also teaches that the disclosed methods of genetic improvement are carried out in a line or breed at the level of the genetic nucleus (aka, nucleus breeding farms).” Pet. 29 (citing Ex. 1004, 391–392, 394).

Petitioner asserts that “Dekkers expressly teaches that ‘nucleus breeding farms typically maintain specific breeds or breeding lines’ where ‘specific lines are developed to advance certain traits of interest, while keeping up the main breed characteristics.’ [] This is genetic progress.” Pet. Reply 8–9 (citing Ex. 1004, 394; Ex. 1097 ¶¶ 27–28).

Patent Owner argues that the method steps of claim 1 are not disclosed or rendered obvious in light of Dekkers. *See* PO. Resp. 35–43. Patent Owner argues that the “Petition does not identify any disclosure in the cited references that explains why skewing the swine sex ratio within the genetic nucleus would lead to genetic progress.” *Id.* at 35; *see id.* at 31 (“Noticeably absent from this statement is any discussion of using sexed semen within a single line in the genetic nucleus to create genetic progress.”). Patent Owner argues that genomic selection technology is not sex-selection and, therefore, Dekkers teaches a different method of increasing selection intensity. Sur-reply 17.

Dekkers begins by explaining that “[g]enetic improvement in pigs began several centuries ago leading to a clear transformation from the wild pig to the domestic pig. . . . This was followed by breeding for specific characters and the development of specialized breeds and lines.” Ex. 1004,

390. Dekkers discloses that the “objective of a breeding programme is to produce the ‘most improvement per unit of time.’” *Id.* at 391. Dekkers discloses that “[t]he nucleus farms [in a pyramid scheme] actually conduct breeding and selection for the genetic improvement of specific breeds or lines.” *Id.* at 393–394; Ex. 1002 ¶ 101. Dekkers describes that “[g]enetic improvement within sire or dam lines is then targeted towards improvement in specific characteristics that are important to the producers and consumers.” Ex. 1004, 394; Ex. 1002 ¶ 101. Dekkers explains that “[t]he nucleus breeding farms typically maintain specific breeds or breeding lines. . . . Furthermore, within a breed, specific lines are developed to advance certain traits of interest, while keeping up the main breed characteristics.” Ex. 1004, 394. “The choice of breeding goals and successful breeding have led to the genetic improvement of economically important traits such as daily gain, backfat thickness, feed efficiency and litter size.” *Id.* at 398. Dekkers explains that “[m]ost genetic improvement in pigs is based on selection within pure breeds or lines. Within breed selection can be improved with availability of genotype on markers that are associated with the trait(s) of interest.” *Id.* at 410. According to Petitioner’s expert, Dr. Baes, genetic progress is synonymous with genetic improvement. Ex. 1002 ¶ 102.

Applying the claim construction of “genetic nucleus” as set forth above, Petitioner shows by a preponderance of the evidence that Dekkers meets the preamble limitation of increasing genetic progress in a breeding program targeting specific breeds or lines in a nucleus farm which are located at the top of the pyramid. Dekkers teaches “that the breeding objective is actually implemented at the nucleus level, but it is targeted to

the production level and to future markets.” Ex. 1004, 397. Dekkers further teaches

breeding goals are defined separately for sire and dam lines. The sire lines are mainly selected for genetic improvement in production traits such as growth rate, carcass quality, meat quality, feed conversion and conformation. The breeding goal for dam lines typically includes the same production traits, but in addition female reproductive traits, such as litter size, piglet survival, age at puberty, conception rate, number weaned and weaning to oestrus interval.

Ex. 1004, 397. Because Dekkers teaches that breeding goals are defined separately for sire and dam lines one of ordinary skill in the art would understand that having the ability to preselect progeny for sex would reduce costs associated with breeding. *See* Ex. 1002 ¶ 48 (“Preselecting for sex in a swine dam or sire line in the genetic nucleus was recognized as one of the main applications of sexing technology. . . . The use of sex sorted sperm permits breeders to reduce cost associated with the production of progeny of the undesired sex.”); *see also* Ex. 1055, 80 (“The main application by pig producers is for the use of sex-sorting technology in nucleus herds, particularly for producing female lines.”).

Patent Owner argues that Petitioner and the Board both misinterpret Dekkers’s statement that sexed semen can be “useful for the nucleus” tier, incorrectly inferring that this is a disclosure of using sexed semen to create genetic progress within the nucleus. PO Resp. 32. According to Patent Owner, “[b]reeding events involving the nucleus can be aimed at many different goals.” *Id.* (citing Ex. 2005 ¶¶ 34–37). “Crucially, none of these nucleus breeding activities are aimed at creating genetic progress within the nucleus, nor would a [person of ordinary skill in the art] have envisioned

that as the natural use of sexed semen in the nucleus.” *Id.* at 32–33 (citing Ex. 2005 ¶¶ 34–37).

According to Dekkers, “the breeding objective is actually implemented at the nucleus level, but it is targeted to the production level and to future markets.” Ex. 1004, 397. During oral argument Patent Owner’s counsel, Mr. Moffett, explained that:

Genetic progress happens throughout the whole pyramid, and that’s one of the things, one of the issues we take with petitioner. That they want to take the position that every time you see a reference to genetic progress, for example, in Dekkers, well, that’s necessarily happening in the genetic nucleus. And that is not the case, that’s not what our expert said. In fact, their experts acknowledge that as well.

JUDGE POLLOCK: I’m sorry, counsel. I’m a little bit confused. I thought I just heard you say that in Dekkers, genetic progress happens in the whole pyramid. So it happens in the whole pyramid, it includes the nucleus, right?

MR. MOFFETT: We’re not saying, Your Honor, that genetic progress can’t happen in the genetic nucleus. That’s not our position. Our position is that you cannot interpret every reference to genetic progress, to the genetic – as occurring in the genetic nucleus. In other words, every reference they point to as to genetic progress in Dekkers, they [Petitioner] want you to say, well, that happens definitely in the genetic nucleus.

JUDGE POLLOCK: Are you saying that Dekkers does not teach genetic progress within the nucleus?

MR. MOFFETT: I’m not saying that, Your Honor.

...

JUDGE POLLOCK: So does Dekkers teach genetic progress in the nucleus?

MR. MOFFETT: Yes. But what Dekkers is focused on, and what the disclosure they rely on repeatedly throughout Dekkers is

focused on and talking about, is the passing down of the dissemination through the tiers of genetic progress.

Tr. 30:4–31:9. Just because Dekkers may be interested in traits that are beneficial for the production tiers of the breeding pyramid does not detract from Dekker’s teaching that trait information is also applied at the level of the nuclear herd. *See* Ex. 1004, 397 (“the breeding objective is actually implemented at the nucleus level”). After all, the nuclear herd maintains the traits that are desirable in the production tier. Here, Petitioner’s expert testifies that the primary objective of agricultural breeding is to maximize commercially desirable traits while minimizing undesirable traits within the population, including swine populations intended for food production. Ex. 1002 ¶ 37 (citing Ex. 1004, 391–392; Ex. 1008, 445; Ex. 1009, Abstract).

Based upon our review of the arguments and evidence in the whole record, we determine that the preponderance of the evidence supports Petitioner’s position. Specifically, we agree with Petitioner that a person of ordinary skill in the art would have understood that Dekkers teaches making genetic progress throughout the breeding pyramid. Because the breeding pyramid in Dekkers includes the genetic nucleus this would also include making genetic progress within the genetic nucleus as well. Accordingly, we determine that Petitioner has sufficiently shown that Dekkers teaches the preamble limitation of claim 1.

(Element 1a) “providing a boar from a genetic nucleus and a sow from a genetic nucleus”

Petitioner asserts that Dekkers discloses the element of providing a boar from a nuclear herd for the purpose of breeding the animals. *See* Pet. 29–31.

Patent Owner does not dispute that Dekkers teaches mating between boars and sows to generate offspring. *See generally* PO Resp. 24–43.

Based upon our review of the arguments and evidence in the complete record, we determine that Petitioner has sufficiently shown that Dekkers teaches element 1a. *See* Dec. 20–22.

(Element 1b) “collecting a semen sample from said boar”

Petitioner contends that Dekkers teaches sex sorting the semen and artificial insemination. *See* Pet. 32–33. Petitioner contends that one of ordinary skill in the art at the time the invention was made would have understood that, artificial insemination of the sow in a sire line with sperm from a boar in the same line “necessarily requires first collecting the boar sperm from said boar.” Pet. 33 (citing Ex. 1002 ¶ 111).

Patent Owner does not dispute that Dekkers teaches mating between boars and sows to generate offspring. *See generally* PO Resp. 24–43.

Based upon our review of the arguments and evidence in the complete record, we determine that Petitioner has sufficiently shown that Dekkers teaches element 1b. *See* Dec. 22–23.

(Element 1c) “sorting said semen sample into at least two subpopulations of sperm cells, wherein at least 80% of a first subpopulation bears X-chromosomes or Y-chromosomes”

Petitioner asserts that Dekkers teaches sorting boar semen, which means sorting the sample into at least 2 subpopulations of sperm cells where

one subpopulation contains X chromosome and the other subpopulation contains Y chromosomes. Pet. 33–35 (“The ‘separation of spermatozoa,’ as described in Dekkers, by definition, produces at least two subpopulations (e.g., a subpopulation of predominantly X-chromosome bearing sperm cells and another subpopulation of predominantly Y-chromosome bearing sperm cells).”). Petitioner’s expert, Dr. Baes, testifies that one of ordinary skill in the art at the time the invention was made would have understood that “[f]rom a commercial pig production perspective, using semen with less than 80% sex skew (i.e., purity for X or Y chromosomes) would not be efficient or economical, and consumers (e.g., pig producers) would not purchase such poorly enriched samples.” Ex. 1002 ¶ 113; Pet. 34. Petitioner’s expert explains that one of ordinary skill in the art at the time the invention was made in light of the teachings of Dekkers, specifically Figure 16.4 shown above (II.D.1), would have known that in order “to produce females for a gilt line or males for a boar line by selecting a subpopulation of sorted sperm cells containing predominantly X-chromosomes to increase the likelihood of producing female offspring or containing predominantly Y-chromosomes to increase the likelihood of producing male offspring, respectively.” Ex. 1002 ¶ 131 (citing Ex. 1004, 396).

Patent Owner disputes that Dekkers discloses sorting semen samples having “at least 80%” of a population of sperm cells that carry either X-chromosomes or Y-chromosomes. PO Resp. 26 (“This paragraph [in Dekkers] makes only a general reference to ‘sperm sexing technology,’ and is not an express disclosure of sorting at an 80% skew rate (or any other rate)”; Ex. 2005 ¶¶ 58–63; Ex. 2004, 54:17–55:10. Patent Owner argues that there is no reasonable expectation of success in using sexed semen in

commercial pig production. PO Resp. 39 (“none of those references [Dekkers, Maxwell, Garner, and Vazquez] report the successful use of sexed semen in any part of the commercial pig production pyramid, much less for the purpose of improving genetic progress at the nucleus herd level. Indeed, many of Petitioner’s references include express caveats that, for swine, sexed semen has not been implemented outside of a research setting.”).

The issue is what one of ordinary skill in the art at the time the invention was made would have understood about using sex sorted semen in the context of artificial insemination for the purpose of directing piglet production toward creating predominantly male or female offspring.

Dekkers discloses that

sperm sexing technology allow[s] for the production of either male or female offspring for sire or dam lines, and this can be very useful for the nucleus and multiplication tiers. The technique involves the separation of spermatozoa carrying either X or Y chromosomes (Maxwell *et al.*, 2004; Garner, 2006). The sorted sperm can then be used for AI or embryo transfer.

Ex. 1004, 396.

Petitioner’s expert explains that the state of the art at the time the invention was made required at least 80% sex sorted semen to adequately produce predominantly female or male offspring. *See* Ex. 1002 ¶ 113. Specifically, Petitioner’s expert avers that one of ordinary skill in the art would have understood that “[f]rom a commercial pig production perspective, using semen with less than 80% sex skew (i.e., purity for X or Y chromosomes) would not be efficient or economical, and consumers (e.g., pig producers) would not purchase such poorly enriched samples.” Ex. 1002 ¶ 113. Petitioner’s expert testimony is supported by the teachings of

Maxwell (Ex. 1005) and Garner (Ex. 1006), both of which are expressly cited in Dekkers as evidencing “sperm sexing technology . . . for the production of either male or female offspring for sire or dam lines.” *See* Ex. 1004, 396, 420, 422. In particular, Maxwell teaches that “changes in the design of the sorter electronics, [] allowed significant increases in sorting rates (15 million spermatozoa per hour) to be achieved while maintaining more than 85% purity.” Ex. 1005, 81. Gardner teaches that “acceptable sperm sexing efficiency, usually at 90% accuracy.” Ex. 1006, Abstract. Based on these teachings in Maxwell or Garner, we find that the evidence of record supports Petitioner’s contention that sorting at a level of at least 80% would have been obvious.

Patent Owner argues that “to prove obviousness, Petitioner must do more than just show that a [person of ordinary skill in the art] would have a reasonable expectation of successfully sex-selecting pig semen and achieving a pregnancy in a laboratory environment.” PO Resp. 39 (citing *Teva Pharms. USA, Inc. v. Corcept Therapeutics, Inc.*, 18 F.4th 1377, 1381 (Fed. Cir. 2021) (holding that where the claims required administering a specific 600 mg dosage, “the Board was required to frame its reasonable-expectation-of-success analysis around that specific dosage.”). Specifically, Patent Owner argues that Petitioner’s references “include express caveats that, for swine, sexed semen has not been implemented outside of a research setting.” PO Resp. 39. In other words, Patent Owner argues that because sexed semen is not being used “in any part of the commercial pig production pyramid, much less for the purpose of improving genetic progress at the nucleus herd level” it could not render the use of sexed semen obvious. *Id.*

Petitioner responds that Patent Owner “again improperly reads a commercial viability and/or commercial-scale limitation into the ’734 claims.” Pet. Reply. 20. Petitioner contends that even Patent Owner’s expert acknowledges that “the methods employed by the ’734 patent were known in the art at the time of filing.” *Id.* (citing Ex. 1096, 19:18–23, 27:3–8; Ex. 1097 ¶ 64); *see* Ex. 1097 ¶ 61. Petitioner contends that both Petitioner’s and Patent Owner’s experts agree that herds making up the genetic nucleus are much smaller in number than the herds at the commercial production level. *Id.* at 21 (citing Ex. 1097 ¶¶ 65, 84; Ex. 1096, 36:21–37:4). Petitioner contends that sperm sorting combined with low-dose insemination would have been practical for the much smaller herds found at the genetic nucleus. *Id.* at 21; *see* Ex. 1097 ¶ 65 (Dr. Baes testimony that “the relatively smaller herd size of the genetic nucleus (*e.g.*, hundreds or thousands as compared to millions at the production level) makes the use of sexed semen far easier than in vastly larger multiplier and production herds”).

We find that Petitioner has the better position. Here, Patent Owner is identifying the scope of the invention as the commercial tier at the bottom of the pig production pyramid. However, the scope of the *claimed* invention is directed to the substantially smaller herd sitting at the top of the commercial pig breeding pyramid. As explained in *Teva*, “[t]he reasonable-expectation-of-success analysis must be tied to the scope of the claimed invention. *See Allergan, Inc. v. Apotex Inc.*, 754 F.3d 952, 966 (Fed. Cir. 2014) (“[F]ailure to consider the appropriate scope of the ... claimed invention in evaluating the reasonable expectation of success ... constitutes a legal error ...”); *see also Intelligent Bio-Sys., Inc.*, 821 F.3d at 1367.” *Teva*, 18 F.4th at 1381.

Above, we construed “genetic nucleus” as an animal herd comprising one or more lines and positioned at the top of a multi-level pyramidal hierarchy in a commercial herd breeding structure. *See above* II.C.1. The herd at the top of the pyramid comprises far fewer animals than the multiplication and production tiers. *See* Ex. 1097 ¶¶ 65 (“the relatively smaller herd size of the genetic nucleus (e.g., hundreds or thousands as compared to millions at the production level) makes the use of sexed semen far easier than in vastly larger multiplier and production herds”), 84; Ex. 1096, 36:21–37:4; Ex. 2004, 39:6–22 (millions of animals at the production level). Petitioner contends that sperm sorting combined with low-dose insemination would have been practical for the much smaller herds found at the genetic nucleus. Pet. Reply 21. Both Maxwell and Garner reasonably suggest that deep-uterine insemination (DIU) allows for the use of sex sorted boar sperm. *See* Ex. 1005, 88; Ex. 1006, 952. Patent Owner admits that the use of sex-selected semen in the laboratory environment works. *See* PO Resp. 39. Even if the genetic nucleus (i.e. nuclear herd) comprises in excess of 100 or even 1000 animals it is not clear why using artificial insemination in conjunction with sex-selection would not work at the level of the genetic nucleus as argued by Patent Owner. *See* PO Resp. 39 (“sexed semen has not been implemented outside of a research setting”).

Absolute predictability is not necessary to reach an obviousness determination; only a reasonable expectation of success is needed. *Pharma Stem Therapeutic, Inc. v. ViaCell, Inc.*, 491 F.3d 1342, 1360 (2007); *In re Kubin*, 561 F.3d 1351, 1360 (Fed. Cir. 2009); *In re O’Farrell*, 853 F.2d 894, 903–904 (Fed. Cir. 1988) (“Obviousness does not require absolute predictability of success For obviousness under § 103, all that is

required is a reasonable expectation of success.”). In particular, Dekkers teaches that sorted sperm can be used for artificial insemination. Specifically, acknowledging that “new developments in sperm sexing technology allow for the production of either male or female offspring for sire or dam lines, and this can be very useful for the nucleus and multiplication tiers.” Ex. 1004, 396; Ex. 1002 ¶ 48; *see also* Ex. 1055, 80 (“The main application by pig producers is for the use of sex-sorting technology in nucleus herds, particularly for producing female lines.”).

Here, the evidence supports Petitioner’s contentions that given the teachings of Dekkers one of ordinary skill in the art would have appreciated that using sex sorted semen when breeding in a dam line, would increase the number of female offspring, and by increasing the population of females the likelihood of finding a desirable trait associated with females in a particular line is also increased. Ex. 1004, 396 (“sperm sexing technology allow for the production of either male or female offspring for sire or dam lines, and this can be very useful for the nucleus.”); Ex. 1097 ¶¶ 41–44. The same would apply for using the same type of selection in a sire line.

Patent Owner contends that “the Board conceives of a potential explanation—not presented in the Petition—that one would want to use sexed semen to produce females for dam lines and males for sire (or boar) lines.” PO Resp. 36 (citing Paper 8 at 39). Indeed, dam lines and sire lines contain both male and female animals – a fact that is undisputed. *Id.* (citing Ex. 2005, ¶ 20; Ex. 2004, 20:24–21:2).

Petitioner responds:

Inguran invents a flaw not found in the ID [Institution Decision]. EX1097, ¶54. Inguran suggests the Board assumed that dam and

sire lines consist of a single sex only. POR, 37. This is nonsense as the ID merely notes “breeding goals are defined separately for sire and dam lines.” ID, 39.

Pet. Reply 18.

Dekkers teaches “that the breeding objective is actually implemented at the nucleus level, but it is targeted to the production level and to future markets. In commercial breeding, breeding goals are defined separately for sire and dam lines.” Ex. 1004, 397. Dekkers explains that “new developments in sperm sexing technology allow for the production of either male or female offspring for sire or dam lines, and this can be very useful for the nucleus and multiplication tiers.” *Id.* at 396.

Based upon our review of the arguments and evidence in the complete record, we determine that the evidence sufficiently supports Petitioner’s position that using sex sorted semen at a level of 80% or better is reasonably suggested by the combination of Dekkers in conjunction with either Maxwell or Garner meeting the limitation of element 1c.

(Element 1d) “inseminating said sow with sperm cells from said first subpopulation to produce offspring from said sow”

Petitioner asserts that Dekkers discloses the use of sorted sperm for artificial insemination, which a person of ordinary skill in the art would have understood to mean inseminating the sow with the sex sorted semen. Pet. 35 (citing Ex. 1004, 417; Ex. 1002 ¶¶ 115, 116).

Patent Owner does not dispute that Dekkers teaches artificial insemination in order to impregnate sows. *See generally* PO Resp. 24–43.

Based upon our review of the arguments and evidence in the complete record, we determine that Petitioner has sufficiently shown that Dekkers teaches element 1d. *See* Dec. 26–27.

(Element 1e) “genotyping said offspring to identify one or more genetic markers”

Petitioner contends that Dekkers teaches the use of genetic markers for improvements within the nucleus herd. Pet. 36 (citing Ex. 1004, 407–408; Ex. 1002 ¶¶ 117, 118).

Patent Owner does not dispute that Dekkers teaches genotyping offspring. *See generally* PO Resp. 24–43.

Based upon our review of the arguments and evidence in the complete record, we determine that Petitioner has sufficiently shown that Dekkers teaches element 1e. *See* Dec. 27–28.

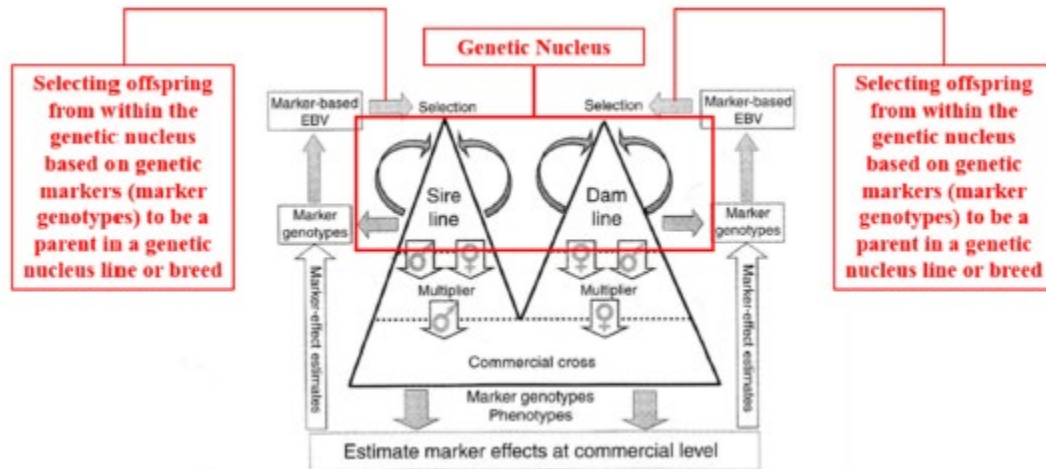
(Element 1f) “selecting one or more of said offspring as a parent in the line or breed in the genetic nucleus based on the one or more identified genetic markers”

Petitioner asserts that Dekkers discloses selecting offspring from the mating of boars and sows in a sire or dam line and using the offspring to further improve the sire or dam line by having the offspring become parents for the next generation in the nucleus herd. Pet. 36–37. In particular, Petitioner asserts that Dekkers discloses

that a sow and boar in the sire line (or the dam line) are bred to produce offspring. The offspring are genotyped to identify genetic markers, as shown by the arrows from the genetic nucleus sire line and dam line, to the boxes titled “Marker genotypes.” An estimated breeding value (“EBV”) is calculated based on genotype results, as indicated by the arrows from the “Marker genotypes” box to the “Marker-based EBV” box. A selection of the offspring to be the parents of the next generation is made based on the EBVs, which were based on the genetic markers.

Pet. 36–37 (citing Ex. 1004, 417 (Fig. 16.4); Ex. 1002, ¶¶ 54, 60, 120, 121).

Petitioner provides an annotated version of Dekkers Figure 16.4, reproduced below:



Petitioner's annotations, shown in red above, identify the genetic nucleus and selection based on marker genotypes and estimated breeding values. Pet. 37 (citing Ex. 1004, 417 (Fig. 16.4)).

Dekkers discloses that “[a] combination of reproduction technologies with advancements in molecular genetics is particularly expected to deliver a powerful impact in increasing the rate of genetic progress and to allow for efficient use of genetic variation and reduce genetic lag.” Ex. 1004, 397. Dekkers teaches the selection of marker genotype of individuals for the purpose of creating commercial cross breeds and to provide feedback to the top of the pyramid. *See above* II.D.1 (reproducing Dekkers Figure 16.4). “[G]enomic selection removes many of the limitations that exist in current phenotype-based programmes with regard to when and on which individuals phenotypes must be evaluated. . . . [G]enomic selection provides for selection within families.” *Id.* at 416. Dekkers teaches that one of the benefits of marker selection is to “increase response to selection for a given rate of inbreeding by increasing selection intensities.” *Id.* Because Dekkers

teaches inbreeding animals, this reasonably suggests the selection of individuals to be part of the genetic nucleus.

Petitioner's expert testifies that the primary objective of agricultural breeding is to maximize commercially desirable traits while minimizing undesirable traits within the population, including swine populations intended for food production. Ex. 1002 ¶ 37 (citing Ex. 1004, 391–392; Ex. 1008, 445; Ex. 1009, Abstract). Petitioner's expert further explains that breeders maintain breeds within a genetic nucleus herd which would require some offspring to be used to perpetuate the line. “Breeds/lines used for the production of females are *dam lines*, and breeds/lines used for production of males are referred to as *sire lines*, respectively.” Ex. 1002 ¶ 40 (citing Ex. 1004, 393–394), *see id.* ¶¶ 42, 107–110.

Here, the identification of estimate breeding values (*see* Ex. 1004, 417 (Figure 16.4)) of individuals in the purebred nucleus populations based on their marker genotypes reasonably supports Petitioner's position that this is “[a] selection of the offspring to be the parents of the next generation is made based on the EBVs, which were based on the genetic markers.” Pet. 37 (citing Ex. 1004, 417 (Fig. 16.4); Ex. 1002 ¶¶ 54, 60, 120, 121); Ex. 1002 ¶¶ 41–42; Ex. 1097 ¶¶ 31–32, 37–40. Patent Owner's expert agrees that this is at least one reasonable interpretation⁵ of Dekkers's Figure 16.4.

Q. And is that marker-based EBV indicated as then being used for the selection process at the top of the pyramid there?

A. Somehow they are able to rank the animals and pick. The eye of the breeder is the best to select the

⁵ Patent Owner's expert during the deposition maintains that Dekkers does not provide a detailed explanation of Figure 16.4. *See* Ex. 1096, 63–78.

animals that would use the next generation.

Q. And the next generation for the genetic nucleus?

...

A. Yes. So, in general, selected animals play a role in the production of the next generation.

Q. I just wanted to clarify. You're talking about for the genetic nucleus?

A. If we interpret the book's sire line as genetic nucleus, then that would be the case.

Ex. 1096, 75:21–76:15.

To summarize, Dekkers teaches that estimated breeding value (EBV) is a way of picking animals for breeding the next generation—including in the next generation in the genetic nucleus. *See* Ex. 1004, 417 (Figure 16.4).

Patent Owner argues that “the Board should not rely on the sweeping, unsupported assertion by Petitioner’s expert that ‘[p]reselecting for sex in a swine dam or sire line in the genetic nucleus was recognized as one of the main applications of sexing technology’ because it can ‘reduce cost associated with the production of progeny of the undesired sex.’” PO Resp. 37 (citing Ex. 1002 ¶ 48).

We are not persuaded by Patent Owner’s contention that Petitioner’s assertion that the use of sex selection reduces costs is unsupported. *See* Ex. 1002 ¶ 48. Petitioner’s expert explains that “[s]exing technology speeds genetic progress by allowing breeders to achieve a desired sex ratio.” *Id.* (citing Ex. 1005, 80). “Preselecting for sex in a swine dam or sire line in the genetic nucleus was recognized as one of the main applications of sexing technology.” *Id.* (citing Ex. 1055, 80 (“The main application by pig producers is for the use of sex-sorting technology in nucleus herds,

particularly for producing female lines.”)). Vazquez, cited by Petitioner’s expert for supporting the position that sex selection reduces costs, further explains:

If pig producers are marketing gilts, the male pigs produced are “by-products.” Consequently, management schemes for livestock production benefit from sex preselection because of the ability to plan matings for a specific sex. Moreover, improved production efficiency of farms has encouraged a reduction in pollutant emissions produced by swine and other domestic animal production facilities []. Thus, a reduced environmental impact could be another important benefit derived from application of sex-sorting technology in swine production.

The discussion about banning castration in male piglets has recently come to the forefront in Europe. Production of females through sex preselection, if it can be done efficiently, will be considered as one alternative to surgical castration. As an added benefit for meat production, consumers in some countries prefer female to male pigs for fresh or cured pork meat.

Ex. 1055, 80–81. Thus, Vazquez supports the position that sex selection can reduce the cost of pig production by reducing the number of “by products.”

Based upon our review of the arguments and evidence in the complete record, we determine that the evidence sufficiently supports Petitioner’s position that a person of ordinary skill in the art would have understood that the combination of Dekkers with either Maxwell or Garner teaches element 1f.

Summary of Claim 1

To summarize, based upon our review of the arguments and evidence on the complete record, we determine that the preponderance of the evidence supports Petitioner’s position that a person of ordinary skill in the art would

have understood that the combination of Dekkers with either Maxwell or Garner would have rendered claim 1 obvious.

Claims 2 and 3

Petitioner asserts that Dekkers discloses a line or breed is a gilt line (claim 2) or a boar line (claim 3). Pet. 38–39.

Patent Owner does not address the limitations of claims 2 and 3 separately, relying instead on the contention that limitations of claim 1 are missing from the combination of Dekkers with either Maxwell, Garner, or Kim. *See generally* PO. Resp. 24–43.

We determine that Petitioner has shown sufficiently that Dekkers teaches the limitation of claims 2 and 3. In particular, Dekkers discloses that breeding at the nucleus level is divided into sire and dam lines.

It is important to note that the breeding objective is actually implemented at the nucleus level, but it is targeted to the production level and to future markets.

In commercial breeding, breeding goals are defined separately for sire and dam lines. The sire lines are mainly selected for genetic improvement in production traits such as growth rate, carcass quality, meat quality, feed conversion and conformation. The breeding goal for dam lines typically includes the same production traits, but in addition female reproductive traits, such as litter size, piglet survival, age at puberty, conception rate, number weaned and weaning to oestrus interval.

Ex. 1004, 397. Dekkers discloses that “new developments in sperm sexing technology allow for the production of either male or female offspring for sire or dam lines, and this can be very useful for the nucleus and multiplication tiers.” Ex. 1004, 396.

Based on the foregoing discussion and our consideration of the record as a whole, we determine that Petitioner has shown persuasively that the

combination of Dekkers in conjunction with either Maxwell or Garner teach each of the elements of claims 2 and 3. Accordingly, we determine that Petitioner has shown by a preponderance of the evidence that claims 2 and 3 would have been obvious based on the combination of Dekkers with either Maxwell or Garner.

Claims 4 and 5

Petitioner asserts that Dekkers discloses genetic markers including single nucleotide polymorphisms (SNPs) (claim 4), that are correlated with feed efficiency, average daily gain, or carcass lean (claim 5). Pet. 39–40.

Patent Owner does not address the limitations of claims 4 and 5 separately, relying instead on the contention that limitations of claim 1 are missing from the combination of Dekkers with either Maxwell, Garner, or Kim. *See generally* PO. Resp. 24–43.

On the complete record, we determine that Petitioner has sufficiently shown that Dekkers teaches the limitations of claims 4 and 5. In particular, Dekkers discloses the use of marker-assisted selection (MAS). Ex. 1004, 407. Dekkers explains that “[o]nce markers associated with the trait have been identified and validated, they can be used for genetic improvement.” *Id.* at 408. “Most genetic improvement in pigs is based on selection within pure breeds or lines. Within-breed selection can be improved with availability of genotype on markers that are associated with the trait(s) of interest.” *Id.* at 410. Dekkers Table 16.6, shown above in § II.E.5.f, discloses a list of known genetic markers, identifying the gene *MC4R* (*melanocortin-4 receptor*) that has been associated with lean growth, fatness, and feed intake traits, and gene *HMGA 1* (*high mobility group AT hook 1*) that has been associated with the backfat thickness trait. *Id.* at 413 (Table 16.6). Dekkers

teaches that single nuclear polymorphism (SNP) panels have become available, which allow for the detection of genetic markers that are associated with the trait based on population-wide linkage disequilibrium markers (LD) and with qualitative trait loci (QTLs). *Id.* at 408.

Petitioner argues that Dekkers expressly describes using SNPs as genetic markers for selection purposes. Pet. 45. Because Dekkers “discloses these [genetic] markers [associated with traits such as lean growth, fatness, feed intake, and backfat thickness] are for use in swine breeding to improve the listed traits, it would have been obvious to select SNPs in Table 16.6 [reproduced above in § II.E.5.f] relating to these traits as genetic markers, including *MC4R*” for selection purposes. Pet. 45. Kim cited by Dekkers teaches that *MCR4* is a SNP and is useful as a genetic marker. In particular, Kim teaches that

given the strong relationship of *MC4R* variants to leanness, growth, and feed intake, this mutation could be used immediately for marker-assisted selection. . . . For instance, in sow lines where appetite is normally decreased after farrowing, selection for the *MC4R* allele 2 could help to improve feed intake. . . . Therefore, genotyping for the *MC4R* mutation in pig breeding lines will improve the selection efficiency of feed-related production traits including growth and leanness.

Ex. 1007, 134.

Based on the foregoing discussion and our consideration of record as a whole, we determine that Petitioner has shown persuasively that the combination of Dekkers in conjunction with either Maxwell or Garner in combination with Kim teach each of the elements of claims 4 and 5. Accordingly, we determine that Petitioner has shown by a preponderance of

the evidence that claims 4 and 5 would have been obvious based on the combination of Dekkers with either Maxwell or Garner, and Kim.

Summary

To summarize, based on the evidence and arguments presented by the parties based on the record as a whole, we determine that Petitioner has sufficiently shown by a preponderance of the evidence that claims 1–5 of the ’734 patent would have been obvious over the combination of Dekkers, Maxwell, Garner, and Kim.

G. Remaining Grounds of Obviousness

In the remaining grounds of obviousness, Petitioner asserts that: (1) claims 1–5 are unpatentable as obvious over Dekkers and Okere, Pet. 46–48, Pet. Reply 24–26; and (2) claims 1–5 are unpatentable as obvious over Buttram, Maxwell, and Kim, Pet. 48–59, Pet. Reply 26–30. Because Petitioner challenges the same claims in these grounds that we conclude are unpatentable based upon the obviousness challenge involving Dekkers in conjunction with Maxwell, Garner, and/or Kim as discussed above, we decline to reach these remaining grounds. *See SAS Inst. Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding a petitioner “is entitled to a final written decision addressing all of the claims it has challenged”); *Boston Sci. Scimed, Inc. v. Cook Grp. Inc.*, Nos. 2019-1594, -1604, -1605, 2020 WL 2071962, at *4 (Fed. Cir. Apr. 30, 2020) (non-precedential) (recognizing that the “Board need not address issues that are not necessary to the resolution of the proceeding” and, thus, agreeing that the Board has “discretion to decline to decide additional instituted grounds once the petitioner has prevailed on all its challenged claims”).

III. CONCLUSION⁶

For the foregoing reasons, we conclude that Petitioner has shown by a preponderance of the evidence that claims 1–5 of the '734 patent are unpatentable as obvious over the combination of Dekkers, Maxwell, Garner, and Kim.

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1–5	102	Dekkers ⁷		1–5
1–5	103	Dekkers, Maxwell, Garner, Kim	1–5	
1–5	103	Dekkers, Okere ⁸		
1–5	103	Buttram, Maxwell ⁹		
Overall Outcome			1–5	

⁶ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

⁷ As evidenced by either Maxwell or Garner (claim 1–4), or Kim (claim 5). See Pet. 26; Pet. Reply 8.

⁸ Petitioner challenges the same claims in this ground as challenged by the combination of Dekkers, Maxwell, Garner, and Kim. For the reasons discussed above, we decline to reach this alternative ground. See *above* II.G.

⁹ Petitioner challenges the same claims in this ground as challenged by the combination of Dekkers, Maxwell, Garner, and Kim. For the reasons discussed above, we decline to reach this alternative ground. See *above* II.G.

IV. ORDER

Accordingly, it is hereby:

ORDERED that based on a preponderance of the evidence claims 1–5 of the '734 patent are unpatentable; and

FURTHER ORDERED because this is a final written decision, the parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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